

TS-7

A 1U analogue tape simulator and compressor: input transformer, high-frequency tape tamer and three compressor modes.



Overview

TS-7 is a rack unit in the spirit of the classic multi-stage tape and transformer boxes. The signal runs through three stages in turn: IRON, an input transformer that thickens and saturates; WARMTH, which tames the top end the way tape compresses its own high frequencies; and COMP, a compressor with three characters.

As on the hardware, the thresholds are fixed: INPUT decides how hard you hit everything, from a little warmth to a full slam. The compressor's make-up gain is automatic, and OUTPUT sets the final level.

Everything non-linear runs oversampled (2x or 4x) to keep the distortion clean of aliasing. The small latency this adds is reported to REAPER, so your tracks stay in time.

Reach for it for

- Gluing a drum or mix bus with tape-style warmth
- Slamming drums for energy and pump (SLAM)
- Thickening bass with transformer saturation
- Taming harsh cymbals and vocals without an EQ (WARMTH)
- Parallel smash compression with MIX

Controls

Main panel

INPUT	The level into the unit, -12 to +24 dB, in 0.5 dB steps. Because the thresholds are fixed, this is the main control: more INPUT means more iron saturation, more warmth and more compression.
COMP	The white key steps through the compressor modes; the lights show the one in use (all dark = off). TX: gentle 2:1 levelling, slow, into a saturating output transformer. BUS: smooth 3:1 programme compression that lets the transients through. SLAM: a feedback compressor with ultra-fast attack and release that slams transients and pumps.
DRIVE meter	The level after INPUT, from -20 to +6 dBFS. Green is comfortable, yellow at 0, red above.
WARMTH meter	How much top end WARMTH is taking, 1 - 10 dB.
COMP meter	The compressor's gain reduction, 1 - 20 dB.
OUTPUT meter	The final output level, -24 to -1 dBFS.
WARMTH	The orange key steps the high-frequency tape tamer through OFF and 1 - 6; the display shows the setting. A side-chain watches the top (above about 3.5 kHz) and, when transients push past the threshold, pulls the top down fast and lets it back slowly. Each step lowers the threshold by 4 dB, so it tames more.
IRON	The input transformer, on by default (the amber light). The low band (below about 120 Hz) saturates first, then the whole signal meets a gentle asymmetric soft clip whose even harmonics grow with level. Off = a clean input stage.
BYPASS	Passes the dry signal through (the red light), delayed to match the oversampling so nothing jumps in time. The switch fades over a few milliseconds.
OUTPUT	The final level, -24 to +12 dB, in 0.5 dB steps.

Drag a knob up or down (Shift for fine), use the mouse wheel, or double-click to reset it.

The bottom strip

PRESET	The ◀ / ▶ keys step through the eight presets; click the display for the list. A preset sets INPUT, OUTPUT, WARMTH, COMP, IRON and MIX, and switches BYPASS off.
OS 2X / OS 4X	The oversampling: the key toggles between 2x and 4x (the default). 4x is cleaner on hard settings and uses a little more CPU and latency.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter holding the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default and flashes red when it clips.
MIX	Parallel compression, 0 - 100%: blends the uncompressed signal (after IRON and WARMTH) under the compressed one. Only works with a COMP mode on.
DRIFT	The iron's power supply is not quite steady: its drive and asymmetry wander slowly and unevenly, 0 - 100% (20% by default). The level stays put; only how hard the transformer saturates moves. Very subtle.

Panel keys

S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW	Switches between the black anodised rack panel and a soft, light-grey flat panel with the same layout.

Presets

Init	Unity INPUT and OUTPUT, IRON on, WARMTH and COMP off.
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Tape Glue Bus	+3 dB in, WARMTH 2, BUS compression.
Drum Slam	+6 dB in, WARMTH 3, SLAM, output pulled down 4 dB.
Vocal Tx	+4 dB in, WARMTH 2, TX levelling.
Warm Master	+2 dB in, WARMTH 1, BUS compression at 60% mix.
Bass Iron	+8 dB into the iron with gentle TX compression, no WARMTH.
Parallel Smash	+8 dB in, WARMTH 2, SLAM blended in at 35%.
Lo-Fi Tape	+12 dB in, WARMTH 6 and SLAM: heavily squashed and dark.

Tips

- Set INPUT by watching the DRIVE meter: around 0 is a healthy hit; pushing into the red brings out the iron and the compressor.
- Turn OUTPUT down as you turn INPUT up to keep the level matched, then compare with BYPASS.
- WARMTH is a quick fix for bright cymbals or sibilant vocals: step it up until the WARMTH meter flickers a few dB on the harsh moments.
- SLAM on a drum room or bus with MIX at 30 - 50% adds energy without losing the attack.

File: gfx-ts7.jsfx